Hazel Chen

434-227-9010 znp3rx@virginia.edu
Charlottesville, VA linkedin.com/in/hazelchens/ Project Portfolio

Education

University of Virginia Expected May 2028
B.S Mechanical Engineering, Electrical Engineering Minor Charlottesville, VA
Focus Areas: Electronics Hardware Design, PCB and Circuit Design, Battery and Power Electronics
Relevant Coursework: Circuits, Signals and Systems, Thermodynamics, Fluid Mechanics

Experience

Virginia Motorsports FSAE Team https://vamotorsports.org/

Electrical Lead 2025 to Present

- Lead the electrical subteam of 20+ members across battery, PCB, and high-voltage (HV) systems, determining architecture decisions and design reviews for the tractive system
- Set technical direction and delegated design ownership across battery, HVC, and HV distribution workstreams, keeping subsystems integrated against a shared timeline
- Designed the tractive-system HVC in Altium, managing precharge, contactor control, and isolated BMS communication
- Built and ran a training curriculum in embedded software, PCB design, DAQ, and EV systems for new team members

Motorsports Engineer 2024 to Present

- Designed and built battery and powertrain systems to support EV prototype development
- Debugged hardware and software problems using oscilloscope, logic analyzer, and other electronic test equipment

Student Researcher Summer 2026
University of Virginia – WEARLab engineering.virginia.edu/labs-groups/wearlab

- Developed portable fNIRS device for monitoring hemoglobin levels in ICU patients to support Takotsubo syndrome research

Maker Consultant, UVA Maker Open Studio and Maker Open Lab mol.as.virginia.edu Jan 2025 to Present
University of Virginia

- Led weekly training sessions on fabrication tools and safety practices
- Maintain and repair 8 FFF 3D printers, Formlabs SLA resin 3D printer, 2 CNC routers

Projects *Portfolio Link*

High Voltage Controller (HVC) design for FSAE EV Racecar

- Designed a custom High Voltage Controller PCB handling tractive-battery precharge/discharge and contactor control
- Integrated an ADBMS6822 dual isoSPI transceiver providing isolated communication to the cell-monitoring module boards
- Validated precharge timing and fault response on the bench with oscilloscope

Benchtop Power Distribution Board: A Flexible Power Solution for Electronics Prototyping and Testing

- Designed a custom PCB that provides VCC, 5V, and 3.3V power outputs through WAGO quick connect terminals
- Integrated custom current sensing circuitry for data collection and validation of low voltage systems
- Implemented individually switched and fused channels with transient voltage and reverse polarity protection

FSAE Tractive Battery Thermal Model: MATLAB Heat-Generation Framework

- Built a MATLAB framework estimating cell heat generation and temperature rise over endurance runs for the tractive battery.
- Computed ohmic (I^2R) heating from the pack's current profile to predict adiabatic cell temperature rise to assess whether active cooling is needed
- Structured the model to accept measured cell parameters, with a plan to replace estimated inputs with characterized data

Battery Pack Fusible Link Simulation

- Simulated nickel strip fuse behavior in battery packs using ANSYS Mechanical with coupled thermal and electrical analysis
- Analyzed nickel link geometry and thickness to determine fusing current and temperature rise.

Skills

PCB and Circuit Design: Altium Designer, KiCad, Multisim, Quartus
CAD and Simulation: SOLIDWORKS, Fusion 360 (CAM), ANSYS, nTopology
Programming and Analysis: MATLAB, Python
Fabrication: FFF and SLA 3D printing (Orca Slicer, PreForm), CNC milling, waterjet cutting, sheet metal forming
Languages: Mandarin Chinese (Fluent), English (Native)